



上海赫立智能机器有限公司

**iHORRY** Shanghai iHorry Machine Co.,Ltd.

地址: 上海市松江区玉谷路66号 赫立元吉总部大楼 电话: 021-69186996 传真: 021-69186997 www.ihorry.com

## AOI&GRR Report

Machine model: LX535iLS-DM

Serial Number: GRR-LX0959165825-2

Manufacturer: Shanghai iHorry Machine Co., Ltd.

Calibrator: Fang Mingwen

Calibration date: 2026/9/7



| GR&R test results                                                                                                                                                                                                                                                             |       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| All detected values for this device are within the acceptable range.                                                                                                                                                                                                          |       |
| GR&R for X offset                                                                                                                                                                                                                                                             | 3.60% |
| GR&R for Y offset                                                                                                                                                                                                                                                             | 7.24% |
| Note:<br>When $GR\&R \leq 10\%$ , it indicates that the detection equipment is precise.<br>When $10\% \leq GR\&R \leq 30\%$ , it indicates that the detection equipment is correct.<br>When $GR\&R \geq 30\%$ , it indicates that the detection equipment is of poor quality. |       |

Auditor: Fang Mingwen

Approver: Shi Xiaoyu



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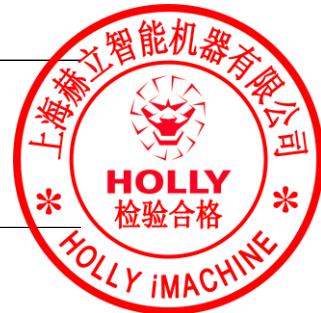
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GR&R - X

| Operator | Trial# | Part     |          |         |         |         |          |          |          |          |          | AVG         |
|----------|--------|----------|----------|---------|---------|---------|----------|----------|----------|----------|----------|-------------|
|          |        | 1        | 2        | 3       | 4       | 5       | 6        | 7        | 8        | 9        | 10       |             |
| OP1      | 1      | -0.0130  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0120  | -0.0030  | -0.0330  | -0.0110  | -0.00640    |
|          | 2      | -0.0140  | -0.0150  | 0.0100  | 0.0000  | 0.0140  | -0.0020  | -0.0120  | -0.0040  | -0.0330  | -0.0130  | -0.00690    |
|          | 3      | -0.0130  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0120  | -0.0030  | -0.0370  | -0.0120  | -0.00690    |
|          | Averag | -0.0133  | -0.0130  | 0.0100  | 0.0000  | 0.0127  | -0.0020  | -0.0120  | -0.0033  | -0.0343  | -0.0120  | XA -0.00673 |
|          | Range  | 0.00100  | 0.00300  | 0.00000 | 0.00000 | 0.00200 | 0.00000  | 0.00000  | 0.00100  | 0.00400  | 0.00200  | RA 0.00130  |
| OP2      | 1      | -0.0130  | -0.0100  | 0.0100  | 0.0020  | 0.0120  | -0.0010  | -0.0120  | -0.0040  | -0.0330  | -0.0110  | -0.00600    |
|          | 2      | -0.0160  | -0.0100  | 0.0100  | 0.0000  | 0.0120  | -0.0010  | -0.0150  | -0.0040  | -0.0330  | -0.0110  | -0.00680    |
|          | 3      | -0.0130  | -0.0100  | 0.0100  | 0.0000  | 0.0120  | -0.0010  | -0.0120  | -0.0040  | -0.0340  | -0.0110  | -0.00630    |
|          | Averag | -0.0140  | -0.0100  | 0.0100  | 0.0007  | 0.0120  | -0.0010  | -0.0130  | -0.0040  | -0.0333  | -0.0110  | XB -0.00637 |
|          | Range  | 0.00300  | 0.00000  | 0.00000 | 0.00200 | 0.00000 | 0.00000  | 0.00300  | 0.00000  | 0.00100  | 0.00000  | RB 0.00090  |
| OP3      | 1      | -0.0150  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0120  | -0.0030  | -0.0330  | -0.0120  | -0.00670    |
|          | 2      | -0.0150  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0120  | -0.0030  | -0.0340  | -0.0110  | -0.00670    |
|          | 3      | -0.0150  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0110  | -0.0030  | -0.0340  | -0.0120  | -0.00670    |
|          | Averag | -0.0150  | -0.0120  | 0.0100  | 0.0000  | 0.0120  | -0.0020  | -0.0117  | -0.0030  | -0.0337  | -0.0117  | XC -0.00670 |
|          | Range  | 0.00000  | 0.00000  | 0.00000 | 0.00000 | 0.00000 | 0.00000  | 0.00100  | 0.00000  | 0.00100  | 0.00100  | RC 0.00030  |
| Part AVG |        | -0.01411 | -0.01167 | 0.01000 | 0.00022 | 0.01222 | -0.00167 | -0.01222 | -0.00344 | -0.03378 | -0.01156 | X -0.00660  |

|                                               |                                  |   |         |
|-----------------------------------------------|----------------------------------|---|---------|
| Rp=                                           | Max(Part AVG) - Min(Part AVG)    | = | 0.04600 |
| R=                                            | (RA + RB + RC) / No of Operators | = | 0.00083 |
| XDIFF=                                        | Max(X) - Min(X)                  | = | 0.00037 |
| UCL=                                          | R * D4                           | = | 0.00215 |
| LCL=                                          | R * D3                           | = | 0.00000 |
| If Trials=3, D4=2.58, D3=0;                   |                                  |   |         |
| If Trials=2, D4=3.27, D3=0, UCL is R's limit, |                                  |   |         |

| Measurement Unit Analysis                |                                           |           |        |        | % Total Variation (TV) |                 |   |
|------------------------------------------|-------------------------------------------|-----------|--------|--------|------------------------|-----------------|---|
| Repeatability-Equipment Variation (EV)   |                                           |           |        |        | EV%                    | = 100 (EV/TV)   |   |
| EV                                       | = R * K1                                  | Trials    |        | K1     |                        |                 |   |
|                                          |                                           | 2         | 0.8862 |        |                        |                 |   |
|                                          |                                           | 3         | 0.5908 |        |                        |                 |   |
| Reproducibility-Appraiser Variation (AV) |                                           |           |        |        | AV%                    | = 100 (AV/TV)   |   |
| AV                                       | = sqrt[ (XDIFF * K2)^2 - (EV^2 / (n*r)) ] | Operators |        | 2      |                        |                 | 3 |
|                                          |                                           | K2        | 0.7071 | 0.5231 |                        |                 |   |
| n = Parts    r = Trials                  |                                           |           |        |        |                        |                 |   |
| Repeatability & Reproducibility          |                                           |           |        |        | GRR%                   | = 100 (GRR/TV)  |   |
| GRR                                      | = sqrt(EV^2 + AV^2)                       | Parts     |        | K3     |                        |                 |   |
|                                          |                                           | 2         | 0.7071 |        |                        |                 |   |
| Part Variation (PV)                      |                                           |           |        |        | PV%                    | = 100 (PV/TV)   |   |
| PV                                       | = Rp * K3                                 | 3         | 0.5231 |        |                        |                 |   |
|                                          |                                           | 4         | 0.4467 |        |                        |                 |   |
|                                          |                                           | 5         | 0.4030 |        |                        |                 |   |
|                                          |                                           | 6         | 0.3742 |        |                        |                 |   |
| Total Variation (TV)                     |                                           |           |        |        | ndc                    | = 1.41 (PV/GRR) |   |
| TV                                       | = sqrt(GRR^2 + PV^2)                      | 7         | 0.3534 |        |                        |                 |   |
|                                          |                                           | 8         | 0.3375 |        |                        |                 |   |
|                                          |                                           | 9         | 0.3249 |        |                        |                 |   |
|                                          |                                           | 10        | 0.3146 |        |                        |                 |   |



赫立经营理念: 勇于创新、锐意进取、诚信合作, 赫立愿与您携手共进、和谐共赢。

## GR&R - Y

|          |        | Part    |          |         |         |         |         |         |         |          |         |    | AVG     |
|----------|--------|---------|----------|---------|---------|---------|---------|---------|---------|----------|---------|----|---------|
| Operator | Trial# | 1       | 2        | 3       | 4       | 5       | 6       | 7       | 8       | 9        | 10      |    |         |
| OP1      | 1      | 0.0050  | -0.0100  | 0.0060  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0250  | -0.0140  | 0.0100  |    | 0.01420 |
|          | 2      | 0.0050  | -0.0100  | 0.0060  | 0.0400  | 0.0300  | 0.0110  | 0.0250  | 0.0250  | -0.0140  | 0.0110  |    | 0.01290 |
|          | 3      | 0.0100  | -0.0120  | 0.0060  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0250  | -0.0140  | 0.0100  |    | 0.01450 |
|          | Averag | 0.0067  | -0.0107  | 0.0060  | 0.0400  | 0.0300  | 0.0117  | 0.0337  | 0.0250  | -0.0140  | 0.0103  | XA | 0.01387 |
|          | Range  | 0.00500 | 0.00200  | 0.00000 | 0.00000 | 0.00000 | 0.00100 | 0.01300 | 0.00000 | 0.00000  | 0.00100 | RA | 0.00220 |
| OP2      | 1      | 0.0050  | -0.0120  | 0.0120  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0270  | -0.0140  | 0.0100  |    | 0.01480 |
|          | 2      | 0.0050  | 0.0000   | 0.0120  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0260  | -0.0140  | 0.0100  |    | 0.01590 |
|          | 3      | 0.0050  | -0.0120  | 0.0120  | 0.0410  | 0.0300  | 0.0120  | 0.0380  | 0.0260  | -0.0150  | 0.0100  |    | 0.01470 |
|          | Averag | 0.0050  | -0.0080  | 0.0120  | 0.0403  | 0.0300  | 0.0120  | 0.0380  | 0.0263  | -0.0143  | 0.0100  | XB | 0.01513 |
|          | Range  | 0.00000 | 0.01200  | 0.00000 | 0.00100 | 0.00000 | 0.00000 | 0.00000 | 0.00100 | 0.00100  | 0.00000 | RB | 0.00150 |
| OP3      | 1      | 0.0050  | -0.0120  | 0.0060  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0260  | -0.0140  | 0.0110  |    | 0.01420 |
|          | 2      | 0.0050  | 0.0000   | 0.0060  | 0.0400  | 0.0300  | 0.0120  | 0.0360  | 0.0260  | -0.0140  | 0.0100  |    | 0.01510 |
|          | 3      | 0.0060  | -0.0120  | 0.0070  | 0.0400  | 0.0300  | 0.0120  | 0.0380  | 0.0260  | -0.0140  | 0.0100  |    | 0.01430 |
|          | Averag | 0.0053  | -0.0080  | 0.0063  | 0.0400  | 0.0300  | 0.0120  | 0.0373  | 0.0260  | -0.0140  | 0.0103  | XC | 0.01453 |
|          | Range  | 0.00100 | 0.01200  | 0.00100 | 0.00000 | 0.00000 | 0.00000 | 0.00200 | 0.00000 | 0.00000  | 0.00100 | RC | 0.00170 |
| Part AVG |        | 0.00567 | -0.00889 | 0.00811 | 0.04011 | 0.03000 | 0.01189 | 0.03633 | 0.02578 | -0.01411 | 0.01022 | X  | 0.01451 |

|                                               |                                  |   |         |
|-----------------------------------------------|----------------------------------|---|---------|
| Rp=                                           | Max(Part AVG) - Min(Part AVG)    | = | 0.05422 |
| R=                                            | (RA + RB + RC) / No of Operators | = | 0.00180 |
| XDIFF=                                        | Max(X) - Min(X)                  | = | 0.00127 |
| UCL=                                          | R * D4                           | = | 0.00464 |
| LCL=                                          | R * D3                           | = | 0.00000 |
| If Trials=3, D4=2.58, D3=0;                   |                                  |   |         |
| If Trials=2, D4=3.27, D3=0. UCL is R's limit, |                                  |   |         |

| Measurement Unit Analysis                 |   |                                                  |        | % Total Variation (TV) |                     |
|-------------------------------------------|---|--------------------------------------------------|--------|------------------------|---------------------|
| Repeatability-Equipment Variation (EV)    |   |                                                  |        |                        |                     |
| EV                                        | = | $R * K1$                                         | Trials | K1                     | EV% = 100 (EV/TV)   |
|                                           | = | 0.00106                                          | 2      | 0.8862                 | = 6.21780%          |
|                                           |   |                                                  | 3      | 0.5908                 |                     |
| Reproducibility-Appraiser Variation       |   |                                                  |        |                        |                     |
| AV                                        | = | $\text{sqrt}[(XDIFF * K2)^2 - (EV^2 / (n * r))]$ |        |                        | AV% = 100 (AV/TV)   |
|                                           | = | 0.00063                                          |        |                        | = 3.70404%          |
| n = Parts                                 |   | r = Trials                                       |        |                        |                     |
|                                           |   | Operators                                        | 2      | 3                      |                     |
|                                           |   | K2                                               | 0.7071 | 0.5231                 |                     |
| Reproducibility-Appraiser Variation (GRR) |   |                                                  |        |                        |                     |
| GRR                                       | = | $\text{sqrt}(EV^2 + AV^2)$                       |        |                        | GRR% = 100 (GRR/TV) |
|                                           | = | 0.00124                                          | Parts  | K3                     | = 7.23747%          |
|                                           |   |                                                  | 2      | 0.7071                 |                     |
|                                           |   |                                                  | 3      | 0.5231                 |                     |
| Part Variation (PV)                       |   |                                                  |        |                        |                     |
| PV                                        | = | $Rp * K3$                                        |        |                        | PV% = 100 (PV/TV)   |
|                                           | = | 0.01706                                          | 4      | 0.4467                 |                     |
|                                           |   |                                                  | 5      | 0.4030                 | = 99.73775%         |
|                                           |   |                                                  | 6      | 0.3742                 |                     |
|                                           |   |                                                  | 7      | 0.3534                 |                     |
|                                           |   |                                                  | 8      | 0.3375                 |                     |
|                                           |   |                                                  | 9      | 0.3249                 |                     |
|                                           |   |                                                  | 10     | 0.3146                 |                     |
| Total Variation (TV)                      |   |                                                  |        |                        |                     |
| TV                                        | = | $\text{sqrt}(GRR^2 + PV^2)$                      |        |                        | ndc = 1.41 (PV/GRR) |
|                                           | = | 0.01710                                          |        |                        | = 19                |

